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**Digital Twin for eHealth: State of the Art and Conceptual
Architecture**

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Abstract

The digital transformation of healthcare has led to the emergence of connected medical devices, electronic health records, remote monitoring platforms, and artificial intelligence-based decision support systems. Although these technologies generate large amounts of heterogeneous data, many eHealth systems still remain fragmented, static, and weakly personalized. Digital Twin technology offers a promising paradigm to address these limitations by creating dynamic virtual representations of physical entities, processes, or patients, continuously updated through data exchange and supported by simulation, analytics, and decision services.

This Master thesis presents a state-of-the-art study of Digital Twins for eHealth. It first reviews the foundations of eHealth, the evolution of the Digital Twin concept, and the distinction between digital models, digital shadows, and fully integrated digital twins. It then analyzes healthcare-oriented Digital Twin applications, architectures, enabling technologies, and implementation challenges. Particular attention is given to interoperability, data governance, privacy, security, explainability, validation, and ethical considerations. Finally, the thesis proposes a conceptual multi-layer architecture for an eHealth Digital Twin system, designed to support continuous monitoring, personalized analysis, scenario simulation, and decision support in a general healthcare context.

Keywords: Digital Twin, eHealth, Digital Health, Healthcare, IoT, IoMT, Artificial Intelligence, Simulation, Interoperability, Clinical Decision Support.

Résumé

La transformation numérique du secteur de la santé a favorisé l'apparition des dispositifs médicaux connectés, des dossiers médicaux électroniques, des plateformes de télésurveillance et des systèmes d'aide à la décision basés sur l'intelligence artificielle. Malgré la quantité importante de données générées, plusieurs systèmes eHealth restent fragmentés, statiques et faiblement personnalisés. La technologie du jumeau numérique constitue une approche prometteuse pour répondre à ces limites, en créant des représentations virtuelles dynamiques d'entités physiques, de processus ou de patients, mises à jour par des échanges de données et enrichies par la simulation, l'analyse et les services décisionnels.

Ce mémoire de Master présente un état de l'art sur les jumeaux numériques appliqués à l'eHealth. Il étudie d'abord les fondements de l'eHealth, l'évolution du concept de jumeau numérique, ainsi que la distinction entre modèle numérique, ombre numérique et jumeau numérique. Ensuite, il analyse les applications, architectures, technologies habilitantes et défis d'implémentation des jumeaux numériques dans le domaine de la santé. Une attention particulière est accordée à l'interopérabilité, à la gouvernance des données, à la confidentialité, à la sécurité, à l'explicabilité, à la validation et aux considérations éthiques. Enfin, le mémoire propose une architecture conceptuelle multicouche pour un système eHealth basé sur un jumeau numérique, capable de soutenir le suivi continu, l'analyse personnalisée, la simulation de scénarios et l'aide à la décision dans un contexte général de santé.

Mots-clés : Jumeau numérique, eHealth, santé numérique, IoT, IoMT, intelligence artificielle, simulation, interopérabilité, aide à la décision clinique.

الملخص

أدت التحولات الرقمية في قطاع الصحة إلى ظهور الأجهزة الطبية المتصلة، والسجلات الصحية الإلكترونية، ومنصات المراقبة عن بعد، وأنظمة دعم القرار المعتمدة على الذكاء الاصطناعي. ورغم أن هذه التقنيات تنتج كميات كبيرة من البيانات المتنوعة، فإن العديد من أنظمة الصحة الإلكترونية ما تزال مجزأة، وثابتة، ومحدودة التخصيص. وتقدم تقنية التوأم الرقمي مقاربة واعدة لمعالجة هذه الحدود، من خلال إنشاء تمثيلات افتراضية ديناميكية للكيانات الفيزيائية أو العمليات أو المرضى، يتم تحديثها باستمرار عبر تبادل البيانات، وتدعيمها بالمحاكاة والتحليل وخدمات دعم القرار.

يعرض هذا البحث مراجعة منظمة لحالة الفن حول توظيف التوائم الرقمية في مجال الصحة الإلكترونية. ويتناول أولاً أسس التوأم الرقمي، بما في ذلك التمييز بين النموذج الرقمي، والظل الرقمي، والتوأم الرقمي، والتوأم الرقمي الذكي، ثم يقدم بيئة الصحة الإلكترونية والصحة الرقمية التي يمكن تطبيق هذه المبادئ داخلها. بعد ذلك، يحلل تطبيقات التوائم الرقمية الموجهة للرعاية الصحية، ومعمارياتها، والتقنيات الممكنة لها، وتحديات تنفيذها. وتستعمل التوائم الرقمية القلبية الوعائية كحالة سريرية ممثلة لأنها تجمع بين الإشارات الحيوية، والتصوير الطبي، والقياسات الهيموديناميكية، والمتابعة الطولية، والنمذجة الخاصة بالمريض. كما يولي البحث اهتماماً خاصاً لقضايا قابلية التشغيل البيئي، وحوكمة البيانات، والخصوصية، والأمن، وقابلية التفسير، والتحقق، والاعتبارات الأخلاقية. وتستخلص مراجعة الأدبيات معمارية عامة متعددة الطبقات للتوائم الرقمية في الصحة الإلكترونية، إضافة إلى مبادئ تتعلق بتدفق البيانات، والمتطلبات العابرة للطبقات، والتقييم. أما الفصل الأخير فيلخص دراسة الماستر ويقدم مشروع CardioTwin كاستمرار عملي في المجال القلبي الوعائي ضمن مشروع نهاية الدراسة، مع الحفاظ على الطابع العام للبحث حول التوائم الرقمية للصحة الإلكترونية.

الكلمات المفتاحية: التوأم الرقمي، الصحة الإلكترونية، الصحة الرقمية، الرعاية الصحية، إنترنت الأشياء، إنترنت الأشياء الطبية، الذكاء الاصطناعي، المحاكاة، قابلية التشغيل البيئي، دعم القرار السريري، التوأم الرقمي القلبي الوعائي.

AI	Artificial Intelligence
API	Application Programming Interface
CDSS	Clinical Decision Support System
CPS	Cyber-Physical System
DL	Deep Learning
DT	Digital Twin
DT4H	Digital Twin for Health
eHealth	Electronic Health
EHR	Electronic Health Record
EMR	Electronic Medical Record
FHIR	Fast Healthcare Interoperability Resources
GDPR	General Data Protection Regulation
HL7	Health Level Seven
ICT	Information and Communication Technology
IoMT	Internet of Medical Things
IoT	Internet of Things
ML	Machine Learning
P4 Medicine	Predictive, Preventive, Personalized and Participatory Medicine
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
SaMD	Software as a Medical Device
TRL	Technology Readiness Level
AF	Atrial Fibrillation
CFD	Computational Fluid Dynamics
CMR	Cardiac Magnetic Resonance
ECG	Electrocardiogram
HF	Heart Failure
LLM	Large Language Model
MRI	Magnetic Resonance Imaging
RHC	Right Heart Catheterization
TTE	Transthoracic Echocardiography
VT	Ventricular Tachycardia

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Part I

General Introduction

CHAPTER 1

GENERAL INTRODUCTION

1.1 Context

Healthcare systems are increasingly supported by digital technologies such as connected medical devices, electronic health records, telemedicine platforms, mobile health applications, and artificial intelligence-based analytics. This evolution has contributed to the emergence of eHealth, a broad domain that aims to improve healthcare delivery, monitoring, prevention, diagnosis, and patient engagement through information and communication technologies.

At the same time, healthcare data are becoming more heterogeneous and continuous. A patient may generate information through clinical consultations, laboratory tests, medical imaging, wearable devices, mobile applications, and hospital information systems. The challenge is no longer only to collect data, but also to transform these data into meaningful, reliable, and actionable knowledge for healthcare professionals and patients.

Digital Twin technology has recently gained attention as a promising paradigm for representing physical entities and processes through dynamic virtual counterparts. In healthcare, a Digital Twin may support the representation of a patient, a medical device, a hospital service, a clinical workflow, or even a population-level process. The objective is to connect real-world data with virtual models in order to monitor current states, simulate possible futures, and support decision-making.

1.2 Problem Statement

Despite the progress of eHealth systems, several limitations remain. First, medical data are often distributed across multiple systems and formats, which makes integration and interoperability difficult. Second, many digital health applications provide static analysis based on isolated measurements rather than dynamic representations that evolve with the real-world situation. Third, decision support systems frequently rely on general population models and may not sufficiently account for individual variability, contextual information, and temporal evolution.

These limitations raise the following central problem:

How can Digital Twin technology be used as a conceptual and architectural foundation for building eHealth systems that are more dynamic, interoperable, personalized, and capable of supporting monitoring, simulation, and decision-making?

1.3 Research Motivation

The motivation of this thesis comes from the need to move from fragmented eHealth applications toward integrated and intelligent healthcare ecosystems. A Digital Twin-based approach can provide a structured way to combine real-world data, virtual representation, simulation, artificial intelligence, and decision services. This is particularly relevant in healthcare because clinical decisions depend not only on current measurements, but also on patient history, context, uncertainty, and possible future trajectories.

1.4 Objectives

The main objective of this thesis is to study the state of the art of Digital Twin technology in eHealth and to propose a general conceptual architecture for eHealth Digital Twin systems. The specific objectives are:

- to review the foundations of eHealth and digital health systems;
- to study the evolution, definitions, components, and maturity levels of Digital Twins;
- to analyze the use of Digital Twins in healthcare from a general eHealth perspective;
- to compare existing architectures, technologies, applications, and limitations;
- to identify major challenges related to interoperability, security, privacy, ethics, and validation;
- to propose a conceptual multi-layer architecture for Digital Twin-based eHealth systems.

1.5 Contributions

The main contributions of this thesis are:

- a structured synthesis of Digital Twin concepts and their relevance to eHealth;
- a classification of healthcare Digital Twin applications from a general perspective;
- a comparative study of selected research papers and architectures;
- a proposed conceptual architecture for a Digital Twin-based eHealth system;
- a discussion of implementation challenges and future research directions.

1.6 Thesis Outline

This thesis is organized into five main parts. Part I introduces the research context, problem statement, objectives, and contributions. Part II presents the background on eHealth and Digital Twin fundamentals. Part III provides the literature review methodology and comparative state of the art. Part IV proposes a conceptual architecture for Digital Twin-based eHealth systems. Finally, Part V concludes the thesis and presents future work.

Part II

Background

CHAPTER 2

EHEALTH AND DIGITAL HEALTH FOUNDATIONS

2.1 Definition of eHealth

eHealth refers to the use of information and communication technologies to support healthcare services, medical decision-making, prevention, diagnosis, monitoring, treatment, administration, and patient engagement. It includes a wide range of systems such as telemedicine, mobile health, electronic health records, connected medical devices, remote monitoring platforms, and clinical decision support systems.

2.2 From Traditional Healthcare to Connected Healthcare

Traditional healthcare is often centered around episodic consultations, paper-based documentation, and hospital-centered workflows. Connected healthcare extends this model by enabling continuous or near-continuous data acquisition, remote communication, and digital coordination among patients, physicians, caregivers, and healthcare institutions.

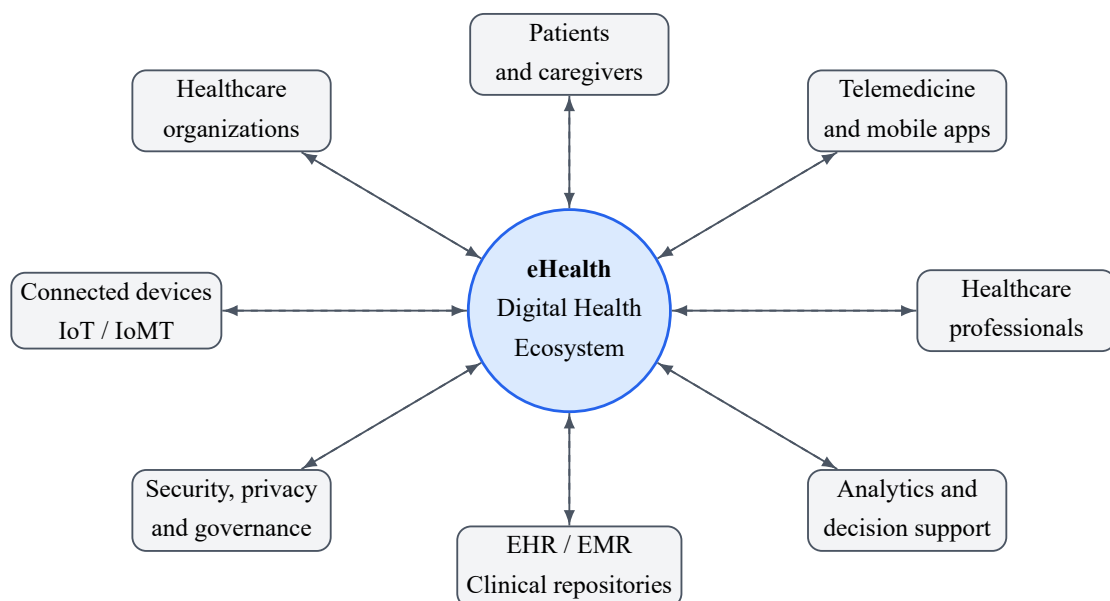


Figure 2.1: General eHealth ecosystem and its main stakeholders.

2.3 Main Components of eHealth Systems

A general eHealth system may include several components:

- **Patient-facing technologies:** mobile applications, wearable sensors, home monitoring devices, and patient portals.
- **Clinical information systems:** EHR, EMR, laboratory information systems, radiology systems, and hospital management platforms.
- **Communication services:** teleconsultation, secure messaging, remote follow-up, and alerts.
- **Analytics and decision support:** machine learning models, rule-based reasoning, risk estimation, dashboards, and report generation.
- **Infrastructure:** cloud platforms, edge devices, APIs, databases, cybersecurity modules, and interoperability standards.

2.4 Data Sources in eHealth

eHealth systems rely on heterogeneous data sources. These include structured clinical data, medical measurements, biosignals, medical images, laboratory results, prescriptions, lifestyle data, environmental data, administrative data, and patient-reported outcomes. The combination of these sources is valuable but challenging due to differences in format, sampling frequency, quality, reliability, and privacy requirements.

2.5 Interoperability Standards

Interoperability is essential for eHealth because patient information may be generated and stored by different systems. Standards such as HL7 and FHIR support the exchange of healthcare information using common formats and resources. In a Digital Twin-based eHealth system, interoperability standards help connect data acquisition modules, clinical repositories, virtual models, and decision services.

2.6 Challenges in eHealth

The main challenges of eHealth include data fragmentation, missing data, inconsistent measurements, limited integration between systems, security and privacy risks, lack of clinical validation, usability barriers, and unequal access to digital services. These challenges justify the need for more integrated, dynamic, and trustworthy approaches such as Digital Twin-based architectures.

2.7 Conclusion

This chapter introduced the foundations of eHealth and highlighted the need for integrated and dynamic systems. The next chapter presents Digital Twin fundamentals and explains how this paradigm differs from conventional digital models and simulations.

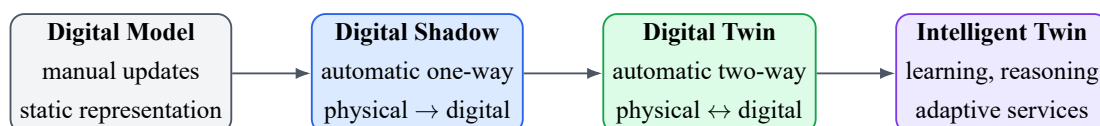
3.1 Origin and Evolution of the Digital Twin Concept

The Digital Twin concept emerged from engineering and product lifecycle management, where virtual representations of physical assets are used to support design, monitoring, prediction, and optimization. Over time, the concept evolved from static digital representations toward dynamic systems connected to real-world data streams and capable of supporting simulation and decision-making.

In its general form, a Digital Twin can be understood as a virtual representation of a physical entity or process that is connected to the real world through data exchange. The virtual representation is not only a visual model; it may include data, models, algorithms, services, and interfaces used to understand and improve the physical counterpart.

3.2 Digital Model, Digital Shadow, and Digital Twin

One important distinction in the literature is between digital models, digital shadows, and digital twins. A digital model is a virtual representation with no automatic data exchange. A digital shadow includes automatic data flow from the physical entity to the digital representation. A full Digital Twin requires stronger integration, where the physical and virtual entities interact through a bidirectional data flow.



Increasing synchronization, autonomy, data integration, and decision-support capability

Figure 3.1: Evolution from digital model to intelligent Digital Twin.

3.3 Core Components of a Digital Twin

Most Digital Twin definitions include three fundamental components: a physical entity, a virtual entity, and a connection between them. More extended views also include data and services as explicit components. In this thesis, the Digital Twin is considered as a system composed of five main elements:

1. **Physical space:** the real-world entity, process, device, patient, or healthcare environment.
2. **Virtual space:** the digital representation that models the state and behavior of the physical counterpart.
3. **Data:** real-time, historical, contextual, and simulated data used to feed and update the twin.
4. **Services:** monitoring, simulation, prediction, optimization, alerts, and decision support.
5. **Connection:** communication mechanisms enabling synchronization between the physical and virtual spaces.

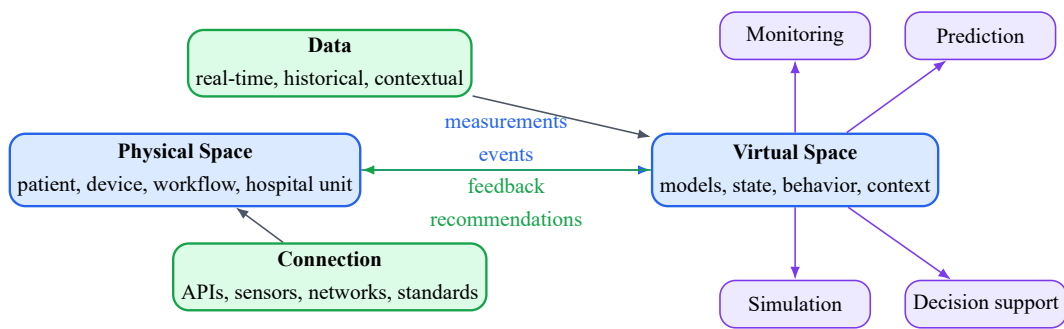


Figure 3.2: Core components of a Digital Twin system.

3.4 Enabling Technologies

Digital Twin systems are enabled by several technologies. The Internet of Things allows physical entities to be monitored through sensors and connected devices. Cloud and edge computing provide infrastructure for data storage and processing. Artificial intelligence and machine learning support pattern recognition, prediction, personalization, and anomaly detection. Simulation models provide interpretable and mechanistic representations of system behavior. Cybersecurity technologies protect data exchange, access control, and system integrity.

3.5 Digital Twin Services

The value of a Digital Twin is expressed through its services. Typical services include real-time monitoring, visualization, anomaly detection, risk estimation, scenario simulation, optimization, predictive maintenance, and decision support. In healthcare, these services must be designed carefully because incorrect outputs may affect clinical decisions and patient safety.

3.6 Digital Twin Maturity

Digital Twins can be classified according to their level of maturity. A low-maturity system may only provide a static representation, while a high-maturity system can learn from real-time data, simulate future behavior, interact with the physical world, and support autonomous or semi-autonomous decisions. In healthcare, fully autonomous Digital Twins remain rare because of clinical safety, ethical, legal, and validation requirements.

3.7 Conclusion

This chapter presented the foundations of Digital Twin technology. The next chapter focuses on how this paradigm is adapted to healthcare and eHealth in a general manner.

4.1 Digital Twin for Health

A Digital Twin for Health is a virtual representation used to support healthcare-related monitoring, simulation, prediction, and decision-making. Depending on its scope, it may represent a patient, an organ, a disease process, a medical device, a clinical workflow, a hospital unit, or a public health system. Unlike a simple dashboard, a Digital Twin aims to maintain a dynamic relationship between real-world data and virtual models.

4.2 Types of Healthcare Digital Twins

Healthcare Digital Twins can be classified into several categories:

- **Patient-specific Digital Twins:** virtual representations of individual patients using multi-modal data.
- **Organ or body-system Digital Twins:** models focusing on a specific organ or physiological system.
- **Disease Digital Twins:** models representing disease progression, risk factors, or treatment response.
- **Medical device Digital Twins:** virtual representations of devices used for design, monitoring, or safety assessment.
- **Hospital and process Digital Twins:** models of workflows, resources, patient flow, logistics, and healthcare units.
- **Public health Digital Twins:** models used for population-level monitoring, planning, prevention, and crisis management.

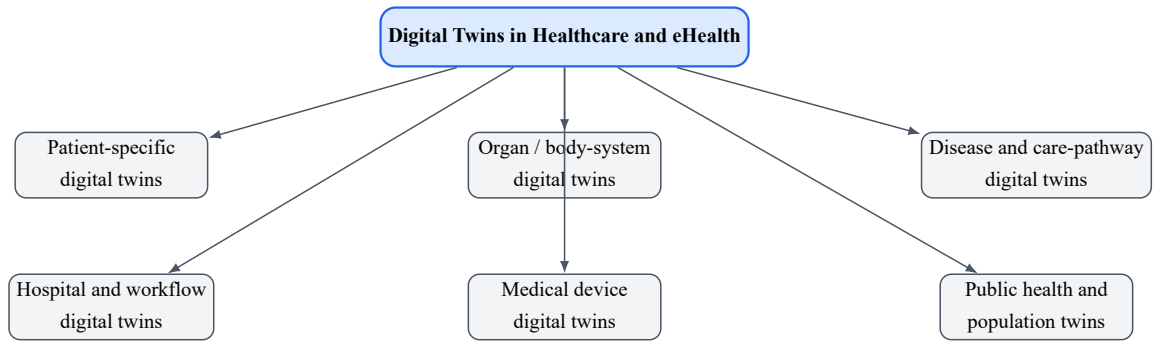


Figure 4.1: General taxonomy of Digital Twin applications in healthcare and eHealth.

4.3 Applications in eHealth

The applications of Digital Twins in eHealth are diverse. They include remote monitoring, personalized care pathways, treatment planning, rehabilitation support, chronic disease management, hospital resource optimization, medical device development, drug discovery, clinical trial simulation, and public health planning. These applications share a common objective: to transform heterogeneous data into actionable knowledge.

4.4 Patient-Centered eHealth Digital Twins

A patient-centered eHealth Digital Twin integrates clinical, physiological, behavioral, and contextual data to support personalized monitoring and decision-making. Such a system may receive information from EHRs, connected devices, laboratory results, patient-reported outcomes, and environmental sources. However, the level of clinical reliability depends on data quality, model validation, and the ability to handle uncertainty.

4.5 Healthcare Process Digital Twins

Digital Twins are not limited to individual patients. They can also represent healthcare processes, such as emergency department flow, outpatient services, operating room scheduling, hospital bed allocation, or vaccination campaigns. In this context, the Digital Twin supports simulation of operational scenarios and optimization of healthcare resources.

4.6 Ethical, Legal, and Social Considerations

Healthcare Digital Twins use sensitive data and may influence medical decisions. Therefore, their development must consider consent, privacy, data ownership, transparency, fairness, bias mitigation, accountability, and clinical responsibility. Ethical design is not an optional component; it is a necessary condition for trustworthy eHealth Digital Twins.

4.7 Challenges and Limitations

The implementation of Digital Twins in eHealth faces several challenges: interoperability between healthcare systems, access to high-quality multimodal data, real-time synchronization, computational cost, cybersecurity, explainability, regulatory approval, integration into clinical workflows, and acceptance by healthcare professionals and patients.

4.8 Conclusion

Digital Twins offer a promising direction for eHealth, but their adoption requires more than technical models. It requires architecture, governance, validation, security, interoperability, and ethical alignment. These issues are addressed in the state-of-the-art chapters that follow.

Part III

State of the Art

5.1 Purpose and Review Design

This thesis uses a structured and focused state-of-the-art review. The objective is not to claim exhaustive coverage of every publication using the term Digital Twin, but to build a sufficiently broad and critically evaluated evidence base for two related domains: Digital Twins in eHealth and healthcare, and Digital Twins in cardiovascular medicine. The first domain establishes general concepts, architectures, services, enabling technologies, and governance requirements. The second provides a clinical deep dive into patient-specific modeling, electrophysiology, hemodynamics, heart failure, treatment simulation, and AI-assisted cardiovascular decision support.

5.2 Research Questions

- **RQ1:** How are Digital Twins defined and distinguished from digital models and digital shadows in healthcare?
- **RQ2:** What patient, organ, device, process, hospital, and public-health applications are reported in eHealth?
- **RQ3:** What architectures, data sources, AI methods, and simulation approaches are used in healthcare Digital Twins?
- **RQ4:** What forms of evidence and validation are reported for healthcare Digital Twins?
- **RQ5:** How are Digital Twins applied to cardiovascular electrophysiology, hemodynamics, heart failure, risk stratification, and treatment planning?
- **RQ6:** What research gaps prevent eHealth and cardiovascular twins from becoming continuously synchronized and clinically deployable systems?

5.3 Source Identification Strategy

Sources were identified through focused keyword searches, publisher and DOI pages, and backward citation tracing from major review papers. Search concepts included “Digital Twin for Health”, “healthcare Digital Twin architecture”, “human Digital Twin”, “AI-powered Digital Twin healthcare”, “cardiovascular Digital Twin”, “cardiac electrophysiology Digital Twin”, “heart failure Digital Twin”, “virtual heart ablation”, and “patient-specific hemodynamic model”. General Digital Twin and AI methodology papers were included only when their concepts were necessary for healthcare interpretation.

5.4 Inclusion and Exclusion Criteria

Studies were included when they provided at least one of the following: a healthcare-DT definition or taxonomy; an architecture; an original patient, organ, process, or public-health application; a cardiovascular patient-specific model with clinical data; a description of AI or hybrid modeling inside a twin; explicit validation evidence; or a substantive analysis of interoperability, uncertainty, ethics, regulation, or governance. Studies were excluded when the term Digital Twin was used only as promotional language, when the physical counterpart and model relationship were unclear, when the work lacked enough methodological detail for comparison, or when it duplicated a more complete source.

5.5 Data Extraction and Comparative Dimensions

For each source, the review extracted the represented physical entity, clinical objective, data modalities, modeling or architecture approach, synchronization direction, maturity, AI role, services, validation evidence, and stated limitations. Cardiovascular studies were additionally analyzed by physiological scale, personalization data, calibration method, mechanistic assumptions, clinical comparator, cohort size when reported, and whether the system was a persistent twin or an offline patient-specific model.

5.6 Synthesis Strategy

The selected studies were synthesized in two states of the art. Within each domain, papers are first discussed thematically and critically, then compared in a dedicated landscape longtable. The synthesis does not rank papers using a single score because the studies pursue different goals and report different evidence. Instead, it identifies convergent architecture components, modeling patterns, maturity limitations, validation gaps, and requirements for future clinical translation.

5.7 Methodological Limitations

This is a structured state-of-the-art review rather than a PRISMA-compliant systematic review. It does not provide database-specific record counts, duplicate-removal statistics, or a meta-analysis. Publication terminology is heterogeneous, and the Digital Twin label is not used consistently. The review therefore applies cautious maturity labels and distinguishes conceptual frameworks, offline simulations, calibrated patient models, digital shadows, process twins, and connected Digital Twins.

5.8 Conclusion

This methodology supports a domain-level analysis of eHealth Digital Twins and a focused clinical analysis of cardiovascular Digital Twins. The following chapter applies the comparison dimensions to both corpora and develops separate comparative tables and research syntheses.

CHAPTER 6

DUAL STATE-OF-THE-ART REVIEW: DIGITAL TWINS IN EHEALTH AND CARDIOVASCULAR MEDICINE

6.1 Chapter Scope and Analytical Approach

This chapter develops two complementary states of the art. The first examines Digital Twins in eHealth and healthcare at the level of patients, organs, medical devices, care processes, hospitals, public health, and precision medicine. The second narrows the analysis to cardiovascular Digital Twins, where patient-specific computational models have been applied to electrophysiology, hemodynamics, heart failure, treatment planning, risk stratification, and virtual testing. The separation is deliberate: the eHealth review identifies the general concepts, architectures, data requirements, services, and governance conditions of healthcare Digital Twins, whereas the cardiovascular review evaluates how these principles are instantiated in one clinically demanding and data-rich domain.

The analysis combines review papers, architecture proposals, methodological studies, and original application studies. Each paper is examined according to the represented physical entity, input data, modeling strategy, physical–virtual synchronization, services, validation evidence, and limitations. A strict maturity interpretation is applied throughout: a patient-specific simulation is not automatically a complete Digital Twin, and strong prediction performance does not replace continuous synchronization, model updating, traceability, or clinical validation. The chapter therefore distinguishes conceptual visions, static models, calibrated patient-specific models, digital shadows, process twins, partial Digital Twins, and more mature connected systems.

6.2 State of the Art I: Digital Twins in eHealth and Healthcare

6.2.1 Evolution of the Healthcare Digital Twin Concept

The transfer of Digital Twin principles from engineering to healthcare introduced a fundamental change in the nature of the physical counterpart. Industrial twins usually represent assets with

relatively well-defined boundaries and operating laws, whereas healthcare twins may represent a changing person, an organ, a disease trajectory, a medical device, a care pathway, or an organizational process. Human physiology is multiscale, partially observable, variable across individuals, and influenced by behavior, treatment, environment, and social context. Consequently, a healthcare Digital Twin cannot be assessed only by geometric fidelity or sensor connectivity; its credibility also depends on clinical meaning, uncertainty, longitudinal consistency, and safe integration into care workflows.

Katsoulakis et al. [1] provide one of the broadest mappings of Digital Twins for Health. Their scoping review organizes applications across hospital management, personalized medicine, surgical planning, medical-device development, drug discovery, in-silico trials, wellness, and public health. The authors also propose the five properties of an impactful health twin: individualized, interconnected, interactive, informative, and impactful. This formulation is valuable because it separates a genuine patient-centered twin from a generic computational model. However, the review also demonstrates that the field remains heterogeneous and that many systems described as Digital Twins lack dynamic bidirectional exchange or real clinical deployment.

Venkatesh et al. [2] frame health Digital Twins as precision-medicine tools created from multimodal patient data, population data, and continuously updated contextual variables. Their contribution is less architectural than translational: they analyze the computational, implementation, and regulatory requirements needed to move from a virtual patient concept toward clinical use. They emphasize that perturbing a digital representation to evaluate possible responses is attractive for precision medicine, clinical trials, and public health, but that validity, fairness, governance, and regulation remain central barriers.

Björnsson et al. [3] present a high-resolution personalized-medicine vision in which an individual model integrates molecular, cellular, clinical, and lifestyle information and is computationally exposed to many candidate treatments. This work is influential because it articulates treatment selection as a patient-specific modeling problem rather than a cohort-average decision. Nevertheless, it is primarily a conceptual and translational framework; the acquisition, synchronization, computational cost, and validation of such high-dimensional patient twins remain unresolved.

Tudor et al. [4] revisit human Digital Twin usage with a definition-based scoping review. Their analysis is especially useful for maturity assessment because it evaluates whether systems called human Digital Twins actually satisfy recognized Digital Twin characteristics. The study indicates that terminology remains inconsistent, application goals vary considerably, and only a subset of reported systems includes the persistent connection and updating expected from a mature twin.

6.2.2 Patient-Specific, Organ-Level, and Process-Oriented Twins

Healthcare Digital Twins can be separated into three broad implementation families. Patient-specific or human twins integrate heterogeneous information to represent an individual state and support personalized monitoring or intervention. Organ-level twins focus on a constrained physiological system and often rely on mechanistic models, imaging, and biosignals. Process-

oriented twins represent hospital units, resource flows, public-health processes, or care pathways and are generally based on event data and discrete-event simulation rather than physiology.

De Benedictis et al. [5] classify healthcare applications and propose a generalized six-layer architecture that separates the physical twin, data, Digital Twin, services, connectivity, and security. Their CanTwin case demonstrates a process-oriented Digital Twin for social-distancing monitoring, queue analysis, people counting, and occupancy supervision. The study is important because it moves beyond a purely conceptual diagram and exposes architectural components in a real organizational setting. Its limitation is equally informative: success in a process twin does not directly establish the validity of a physiological patient twin, because the models, validation criteria, safety consequences, and uncertainty are different.

Noeikham et al. [6] derive a five-layer architecture for a healthcare unit using a structured literature review, software-design patterns, and a chemotherapy-center case study. Their architecture emphasizes modularity, scalability, security, privacy, and integration with Electronic Health Records and Clinical Data Repositories. The work shows that a healthcare-unit twin must connect to existing information systems and that process simulation can evaluate operational changes without disrupting real care. However, the case remains centered on service flow and cannot be generalized directly to patient physiology without additional modeling and clinical-validation layers.

Kamel Boulos and Zhang [7] broaden the scope from personalized medicine to precision public health. Their perspective highlights that Digital Twins can represent populations, places, and health systems as well as individuals. This extension is relevant to eHealth because disease prevention, resource allocation, and emergency response often require population-level simulation. At the same time, the aggregation of individual and environmental data introduces additional privacy, representativeness, and governance concerns.

6.2.3 Artificial Intelligence in Healthcare Digital Twins

Artificial Intelligence is an enabling component of a Digital Twin rather than its defining property. A prediction model may classify disease or forecast risk without maintaining a virtual counterpart of a specific physical entity. Conversely, a synchronized mechanistic or rule-based twin may operate without sophisticated learning. AI becomes especially valuable when the twin must process high-dimensional data, estimate hidden states, personalize parameters, approximate expensive simulations, recognize patterns, or forecast future trajectories.

Janiesch et al. [8] distinguish the broad field of AI from Machine Learning and Deep Learning. Their methodological account is useful for healthcare Digital Twins because it links model choice to data type, sample size, task complexity, computational resources, and interpretability. Conventional ML methods can remain appropriate for structured clinical data and smaller datasets, while deep neural networks are often better suited to images, biosignals, and complex temporal data. The most complex model is not automatically the best clinical model; out-of-sample performance, calibration, explanation, and operational constraints are equally important.

Chaparro-Cárdenas et al. [9] review the convergence of Digital Twins and AI for personalized and predictive healthcare. The paper covers AI-supported diagnosis, rehabilitation, multi-

modal fusion, predictive modeling, and adaptive services, while also stressing privacy, data heterogeneity, edge computing, federated learning, scalability, and clinical validation. A key contribution is the identification of the gap between technically impressive prototypes and systems capable of reliable real-time integration with clinical environments. The review supports the use of hybrid and multimodal intelligence but does not provide evidence that these approaches are already routine clinical practice.

Elgammal et al. [10] analyze 17 recent AI-powered Digital Twin applications across heart health, diabetes, mental health, respiratory care, stress management, and other domains. Their review shows that current systems employ conventional ML, deep learning, generative AI, Large Language Models, and graph-based methods for prediction, personalization, patient-trajectory analysis, and user interaction. It also reveals a maturity problem: many applications are retrospective, experimental, or based on synthetic data, and their clinical workflow integration, uncertainty, external validation, and regulatory status remain limited.

Sharma and Kaur [11] propose a hybrid patient-specific framework combining mechanistic physiological simulation, neural encoders, multimodal clinical data, Bayesian reasoning, and reinforcement learning. The framework directly addresses a central limitation of pure AI systems by grounding prediction in physiological structure. It also addresses limitations of purely mechanistic models by using learning for feature extraction and personalization. Nevertheless, the breadth of the proposed architecture exceeds the clinical evidence currently available; independent validation, prospective testing, and transparent reporting of data and model provenance are required before the approach can be considered clinically actionable.

Across these studies, AI contributes at several stages of the twin lifecycle: artifact detection and preprocessing, latent-state estimation, image segmentation, anomaly detection, diagnosis support, prognosis, patient stratification, parameter calibration, surrogate modeling, scenario ranking, and continuous adaptation. Each role creates a different validation obligation. An imputation model must not silently create clinically important values; a diagnostic model requires discriminative and calibration testing; an adaptive model requires drift monitoring and controlled revalidation; and a surrogate simulation requires evidence that it preserves the behavior of the mechanistic model inside the intended domain.

6.2.4 Interoperability, Validation, and Governance

Interoperability is a structural requirement rather than a peripheral feature. Patient twins may receive data from sensors, mobile applications, EHRs, imaging platforms, laboratories, and clinician inputs, each with different identifiers, formats, semantics, time scales, and data-quality characteristics. Healthcare standards such as HL7 FHIR can facilitate exchange, but syntactic compatibility alone does not guarantee semantic consistency or temporal alignment. The twin must preserve provenance, measurement units, timestamps, uncertainty, and the distinction between measured, calculated, estimated, predicted, and simulated values.

Meijer et al. [12] focus on methodological challenges and show that healthcare Digital Twins require explicit attention to causal assumptions, model updating, uncertainty, missing data, and validation. Their work is particularly valuable because it challenges the tendency to treat a

high-performing predictor as a complete patient twin. They argue for methodological clarity regarding the target represented, the intervention modeled, and the evidence required to support individual-level claims.

Bruynseels et al. [13] analyze the ethical consequences of representing individuals through detailed, longitudinal virtual counterparts. They show that Digital Twins may alter the interpretation of health, normality, prevention, and enhancement by comparing a patient both with population patterns and with their own historical baseline. This creates opportunities for personalized prevention but also risks discrimination, unequal access, excessive surveillance, and opaque secondary use of data. Governance, consent, and transparency must therefore be embedded throughout the architecture.

Clinical credibility requires multiple forms of evaluation. Technical verification asks whether software, data pipelines, and numerical implementations behave as specified. Model validation examines whether outputs reproduce relevant physiological or operational behavior. External validation tests transportability to other populations, institutions, devices, and time periods. Prospective clinical validation evaluates whether the twin improves decisions or outcomes in real workflows. Human-factors evaluation assesses usability, alert burden, trust, and the effect of the system on clinical responsibilities. Most reviewed studies address only a subset of these layers.

6.2.5 Comparative Table for Digital Twins in eHealth

Table 6.1 compares representative eHealth and healthcare Digital Twin studies. The table separates conceptual contribution from implementation evidence and applies cautious maturity labels.

Table 6.1: Comparative state of the art of Digital Twins in eHealth and healthcare.

Study	Scope / represented entity	Data and integration	Modeling / architecture	Maturity	Contribution and evidence	Main limitations
Katsoulakis et al. [1]	Field-wide DT for Health review: patients, organs, hospitals, devices, trials, wellness, and public health	Clinical, molecular, imaging, sensor, environmental, and social data	Scoping review and 5I characterization of healthcare twins	Field-level synthesis	Broad taxonomy of applications and opportunities; distinguishes computational models from health twins	Heterogeneous terminology; many applications remain conceptual; limited standardization and clinical validation
Venkatesh et al. [2]	Precision medicine, population clinical trials, and public health	Multimodal patient, population, environmental, and real-time data	Computational and regulatory perspective	Health-DT vision	Connects patient perturbation and simulation with implementation and regulation	No deployed system; uncertainty, fairness, governance, and regulation remain open
Björnsson et al. [3]	High-resolution individual patient model for treatment selection	Molecular, cellular, clinical, and lifestyle information	Systems medicine and large-scale virtual treatment testing	Conceptual patient twin	Influential personalized-medicine vision linking integrated patient models to drug selection	Very high data and modeling requirements; limited evidence for continuous synchronization and clinical workflow use
Tudor et al. [4]	Human Digital Twins across healthcare applications	Varies by application; clinical, sensor, imaging, and behavioral data	Definition-based scoping review	Maturity audit	Evaluates whether reported human twins comply with recognized DT characteristics	Reveals inconsistent terminology and limited numbers of fully connected, persistent twins
De Benedictis et al. [5]	Healthcare processes and public-health-oriented workplace monitoring	Organizational, location, occupancy, and process data	Generalized six-layer architecture and CanTwin case	Process Digital Twin	Provides explicit functional layers and a real case for social-distancing monitoring	Process-oriented; limited transfer to patient physiology, clinical outcomes, and patient-specific validation
Noeikham et al. [6]	Healthcare-unit service system and chemotherapy center	EHR, CDR, event, resource, and workflow data	Five-layer healthcare-unit architecture and discrete-event simulation	Process Digital Twin	Derives modular architecture with privacy, security, interoperability, and case analysis	Single unit-level case; physiological twinning and broader deployment remain untested
Kamel Boulos and Zhang [7]	Personalized medicine and precision public health	Individual, population, geospatial, and contextual data	Conceptual synthesis of patient and population twins	Conceptual framework	Extends DT thinking from individuals to population planning and public health	Limited implementation evidence; aggregation introduces privacy, bias, and representativeness issues
Meijer et al. [12]	Methodological foundations of healthcare twins	Longitudinal clinical and patient-specific data	Statistical, causal, and validation perspective	Methodological foundation	Clarifies causal assumptions, updating, uncertainty, and individual-level validity	Does not implement a specific twin; recommendations require operational benchmarks

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Table 6.1: Comparative state of the art of Digital Twins in eHealth and healthcare (continued).

Study	Scope / represented entity	Data and integration	Modeling / architecture	Maturity	Contribution and evidence	Main limitations
Chaparro-Cárdenas et al. [9]	AI-enhanced DTs for personalized and predictive healthcare	Biosignals, imaging, EHR, sensors, biomechanical and multimodal streams	Technological review of AI, hybrid reasoning, edge, cloud, and federated approaches	AI-enabled DT synthesis	Maps AI roles in prediction, diagnosis, rehabilitation, personalization, and adaptive services	Data heterogeneity, privacy, scalability, real-time integration, regulation, and controlled clinical testing
Elgammal et al. [10]	AI-powered applications across multiple health domains	EHR, wearables, glucose, imaging, behavioral, respiratory, and population data	ML, DL, generative AI, LLM, and graph-based methods	Review of practical and emerging applications	Identifies 17 applications across heart health, diabetes, mental health, respiratory care, and stress	Many studies remain retrospective or experimental; black-box behavior, interoperability, validation, and adoption barriers
Sharma and Kaur [11]	Patient-specific personalized healthcare	EHR, ECG, imaging, wearable, and clinical data	Hybrid mechanistic simulation, neural encoders, Bayesian inference, and reinforcement learning	Intelligent-DT framework	Proposes continuous synchronization, multimodal fusion, trajectory prediction, and therapy simulation	Broad claims require independent reproduction, external validation, prospective testing, and regulatory assessment
Bruynseels et al. [13]	Ethical implications of human and health twins	Longitudinal biological, physiological, and lifestyle data	Ethical and conceptual analysis	Governance foundation	Identifies implications for normality, prevention, enhancement, privacy, inequality, and discrimination	No technical implementation; ethical principles must be translated into enforceable architecture and policy

6.2.6 Synthesis of the eHealth State of the Art

The eHealth literature shows a progression from broad personalized-medicine visions toward more explicit architectures and application-specific prototypes. Yet the progression is uneven. Process twins currently offer some of the clearest examples of real connectivity because event streams and service metrics can update operational simulations. Patient twins are scientifically more ambitious: they must reconcile incomplete multimodal measurements with models of a biological system that changes over time and cannot be fully observed.

Five conclusions emerge. First, a healthcare twin must have a clearly defined physical counterpart and clinical purpose. Second, longitudinal data and model updating are essential for claims of personalization. Third, AI improves data interpretation and adaptation but does not establish Digital Twin maturity by itself. Fourth, interoperability, provenance, uncertainty, and governance must cross all architecture layers. Fifth, the evidence base remains dominated by reviews, conceptual frameworks, retrospective analyses, and focused prototypes; prospective demonstrations of clinical benefit are comparatively rare.

6.3 State of the Art II: Cardiovascular Digital Twins

6.3.1 From Precision Cardiology to Cardiac Digital Twins

Cardiovascular medicine is a representative Digital Twin domain because cardiac function emerges from coupled electrical, mechanical, fluid, anatomical, and autonomic processes. Clinical data are correspondingly multimodal: ECG describes surface electrical activity, imaging reveals anatomy and motion, catheterization measures pressures and flows, laboratory tests characterize biological state, and wearables provide longitudinal rhythm and activity data. The same diagnosis may arise from different mechanisms, which creates a strong rationale for patient-specific modeling.

Corral-Acero et al. [14] formulate the Digital Twin as an enabling concept for precision cardiology. They organize the field around statistical, mechanistic, and hybrid models and explain how patient-specific representations could support diagnosis, prognosis, device selection, and treatment simulation. Their paper is foundational because it links clinical data streams to multi-scale cardiovascular models. It is also primarily a vision and agenda: many examples available at the time were personalized simulations rather than continuously synchronized twins.

Coorey et al. [15] conduct a mapping review spanning academic publications, commercial activity, and patents. They identify cardiovascular research as a major component of health Digital Twin activity but conclude that most reported applications are numerical simulation models or precursors of the real-time cyber-physical system implied by a strict Digital Twin definition. The review emphasizes precise phenotyping, multimodal integration, prediction, and treatment optimization, while identifying ethical constraints and clinical adoption barriers.

Sel et al. [16] examine the path from cardiovascular Digital Twin principles to clinical impact. Their analysis highlights multimodal sensing, mathematical models, patient-specific calibration, machine learning, and uncertainty. The paper is particularly useful for distinguishing a persistent individual representation from a one-time personalized model and for stressing that

trustworthy cardiovascular twins require bias control, validation, and interpretable estimates.

Cluitmans et al. [17] focus on cardiac electrophysiology. They review applications in sudden-death risk, atrial fibrillation, ventricular arrhythmia, and rhythm-control therapy, while emphasizing the technical difficulty of calibration and the challenge of maintaining a functionally equivalent twin as the patient's condition changes. Their review shows that electrophysiology is among the most developed cardiac-twinning areas but still depends heavily on offline imaging, inverse problems, and computational modeling.

Bhagirath et al. [18] provide a complementary state-of-the-future review centered on cardiac electrophysiology. They distinguish anatomical personalization from mechanistic and functionally equivalent twinning, examine calibration and data-assimilation requirements, and review applications in pacing, cardiac resynchronization, atrial fibrillation, and ventricular tachycardia. Their analysis is especially valuable because it warns against treating every personalized model as a complete twin: the physiological attributes reproduced, the parameters calibrated, and the unmodeled aspects of the patient must be stated explicitly. The review also identifies continuous updating, automated model-generation pipelines, multimodal integration, wearable data, and credible clinical trials as prerequisites for translation.

6.3.2 Patient-Specific Electrophysiology and Arrhythmia Risk

Several influential studies predate widespread use of the Digital Twin label but provide essential technical foundations. Arevalo et al. [19] construct personalized three-dimensional post-infarction heart models from MRI and simulate arrhythmia inducibility. Their proof-of-concept study demonstrates that mechanistic patient-specific modeling can improve risk stratification beyond conventional clinical markers. The approach is an important precursor to cardiac Digital Twins, although it is an offline virtual-heart test rather than a continuously updated twin.

Prakosa et al. [20] extend personalized virtual-heart modeling to infarct-related ventricular tachycardia ablation. The study includes retrospective animal and human analyses and a small prospective feasibility component. Patient-specific imaging is converted into electrophysiological models used to identify potential ablation targets. The work provides stronger clinical translation than a purely retrospective simulation, but the modeling pipeline remains computationally intensive and is not a persistent longitudinal twin.

Zhang et al. [21] develop genotype-specific heart Digital Twins for arrhythmogenic right ventricular cardiomyopathy. The framework integrates contrast-enhanced MRI-derived structural remodeling with genotype-specific cellular electrophysiology. In 16 retrospectively studied patients, the twins predicted ventricular-tachycardia circuit locations with high reported sensitivity, specificity, and accuracy relative to clinical electrophysiology studies. The contribution is important because genotype information is incorporated into patient-specific mechanistic models. The cohort is nevertheless small, and prospective multi-center validation is needed.

Hwang et al. [22] construct patient-specific left-atrial models from computed tomography and electroanatomical maps to perform a virtual amiodarone test after atrial-fibrillation ablation. Virtual response to drug concentrations was associated with one-year recurrence outcomes, with lower recurrence in the group whose virtual AF terminated at therapeutic doses. This study

illustrates a clinically meaningful treatment-simulation service and links virtual predictions to follow-up outcomes. Its limitations include retrospective design, disease- and drug-specific assumptions, and the need for external prospective replication.

O’Hara et al. [23] investigate how cardiac magnetic-resonance resolution affects Digital-Twin predictions of post-infarction ventricular-tachycardia circuits. For six patients undergoing ablation, the authors reconstructed three twins per patient from clinical-resolution and over-sampled late-gadolinium-enhanced MRI and compared predicted circuits with intraprocedural electroanatomical mapping. Higher-resolution twins reduced partial-volume effects and identified a larger proportion of unique circuits, including an emergent circuit after an unsuccessful initial ablation. The study demonstrates that imaging resolution is not a neutral preprocessing choice: it changes scar geometry, predicted re-entry pathways, and potential treatment targets. Its small cohort and preprocedural offline workflow prevent conclusions about general clinical effectiveness, but the work provides concrete validation against procedural data.

6.3.3 Whole-Heart, Electromechanical, and Hemodynamic Models

Gerach et al. [24] present a fully coupled multiscale whole-heart framework integrating electrophysiology, excitation–contraction coupling, mechanics, and closed-loop circulation. The study provides a detailed mechanistic foundation and discusses adaptation to patient-specific measurements. Its strength is physiological coverage; its limitation is computational and calibration complexity. A high-fidelity whole-heart model can support explanation and scenario testing, but without routine updating from an individual patient it remains a platform for building twins rather than a deployed twin itself.

Viola et al. [25] address computational feasibility using GPU acceleration. Their multi-physics model reproduces a heartbeat within hours and is demonstrated for left bundle branch block and cardiac resynchronization. This creates opportunities for synthetic cohorts and repeated simulation campaigns that would be impractical with slower solvers. However, simulation speed alone does not ensure patient-specific validity, and the proof-of-concept is not evidence of real-time clinical synchronization.

Salvador et al. [26] address the same computational bottleneck through a physics-informed AI surrogate. Their Latent Neural Ordinary Differential Equation model learns pressure–volume dynamics from 400 high-fidelity electromechanical simulations spanning 43 parameters related to electrophysiology, mechanics, and cardiovascular hemodynamics. The surrogate enables global sensitivity analysis, parameter estimation, and uncertainty quantification on a single processor within a few hours, illustrating how learned reduced-order models can make repeated whole-heart analyses more tractable. The contribution is methodological rather than a deployed clinical twin: training data originate from simulations of a heart-failure case, and robustness beyond the modeled parameter domain requires further patient-level validation.

Geddes et al. [27] develop patient-specific pulmonary-artery models for heart-failure monitoring. Computed-tomography pulmonary angiography is used to reconstruct anatomy, while invasive hemodynamic measurements provide boundary information and validation targets. The study investigates the minimum geometric complexity needed to predict pulmonary artery pres-

sure and validates predictions against implantable-monitor measurements in a proof-of-concept cohort. The work is clinically relevant because it aims to derive predictive hemodynamic markers noninvasively. Its limitations include dependence on imaging quality, boundary-condition assumptions, computational workflow, and a small validation cohort.

6.3.4 AI-Enhanced Cardiovascular Twins and Population-Scale Modeling

Martinez-Velazquez et al. [28] propose Cardio Twin, an edge-oriented architecture for ischemic-heart-disease detection. The work is an early example of combining connected data, classification, and edge processing under the Digital Twin label. It demonstrates that resource-constrained devices can support parts of a cardiac-twin pipeline, but its virtual heart representation and physiological simulation are limited compared with later mechanistic frameworks.

Gu et al. [29] create patient-specific mechanistic cardiovascular models for 343 heart-failure patients. The optimized parameters and corresponding simulations are treated as Digital Twin features and used in unsupervised phenotyping and supervised prognosis. The integration of physiological prior knowledge with AI improves interpretability and transferability compared with models using clinical variables alone. The study is a strong example of model-informed AI, but measurements were assembled retrospectively, sometimes within a multi-month window, and the twins were not continuously synchronized in real time.

Qian et al. [30] demonstrate population-scale cardiac twinning by constructing 3,461 cardiac Digital Twins from UK Biobank data and 359 additional twins from an ischemic-heart-disease cohort. Cardiac MRI and ECG are used to infer electrophysiological properties, revealing associations between conduction, repolarization, anatomy, age, obesity, lifestyle, and outcomes. The scale is a major contribution because traditional high-fidelity personalization is usually limited to small cohorts. However, the resulting twins are derived from episodic research data and are not persistent clinical replicas that evolve continuously with each participant.

Recent AI surrogate approaches aim to reduce the cost of whole-heart models by learning approximations to high-fidelity simulations. These methods can enable rapid parameter estimation, sensitivity analysis, and near-real-time scenario evaluation. Their clinical credibility depends on domain-of-validity analysis, uncertainty quantification, and safeguards against extrapolation. The hybrid strategy is promising because the mechanistic model defines physiological structure while AI accelerates inference, but it also introduces two interacting error sources: the mechanistic assumptions and the learned surrogate approximation.

6.3.5 Comparative Table for Cardiovascular Digital Twins

Table 6.2 compares the principal cardiovascular studies according to clinical objective, data, modeling, evidence, and maturity. Foundational personalized-heart studies are included and explicitly labeled as precursors where they do not implement persistent synchronization.

Table 6.2: Comparative state of the art of cardiovascular Digital Twin and patient-specific virtual-heart studies.

Study	Clinical objective / entity	Patient data	Modeling approach	Maturity	Evidence / main contribution	Main limitations
Corral-Acero et al. [14]	Precision cardiology across diagnosis, prognosis, and treatment	Imaging, biosignals, clinical, molecular, and device data	Statistical, mechanistic, and hybrid modeling roadmap	Field vision	Establishes the multiscale precision-cardiology DT agenda and application taxonomy	Position paper; limited direct implementation and clinical outcome evidence
Coorey et al. [15]	Mapping cardiovascular DT research, industry, and patents	Clinical, imaging, molecular, device, and contextual data	Mapping review	Emerging field synthesis	Identifies academic, commercial, and patent activity and clarifies potential CVD applications	Most studies are simulations or precursor models rather than synchronized clinical twins
Sel et al. [16]	Principles and clinical impact of cardiovascular twins	Multimodal sensors, imaging, clinical variables, and physiological signals	Review of mathematical, AI, sensing, and validation methods	Translational framework	Connects personalization, persistent representation, uncertainty, and clinical impact	Implementation evidence remains fragmented; bias, VVUQ, updating, and workflow integration are open
Cluitmans et al. [17]	Cardiac electrophysiology, arrhythmia risk, and rhythm-control therapy	ECG, imaging, electroanatomical and electrophysiological data	State-of-the-art review of mechanistic EP twins	Domain review	Synthesizes sudden-death, AF, and VT applications and calibration challenges	Routine clinical deployment and dynamic longitudinal updating remain limited
Bhagirath et al. [18]	Mechanistic Digital Twins for cardiac electrophysiology and precision treatment	MRI, CT, ECG, EHR, smart-watch, and electrophysiological data	State-of-the-future review of anatomical, mechanistic, and functionally equivalent twins	Domain review / research agenda	Defines personalization levels and analyzes calibration, pacing, CRT, AF, VT, and translation requirements	Clinical success stories and continuously updated functionally equivalent twins remain rare
Arevalo et al. [19]	Sudden-death risk after myocardial infarction	Patient MRI and clinical outcomes	Personalized 3D electrophysiological heart models and virtual arrhythmia induction	Offline patient-specific model / DT precursor	Proof-of-concept risk stratification outperformed several conventional markers	Retrospective cohort; no continuous synchronization; computational and modeling assumptions
Prakosa et al. [20]	Guidance of infarct-related VT ablation	Cardiac imaging and electrophysiological information	Personalized virtual-heart simulation of arrhythmogenic substrate	Patient-specific intervention model	Retrospective animal/human analyses plus small prospective feasibility component	Small prospective sample; complex pipeline; not a persistent longitudinal twin
Martinez-Velazquez et al. [28]	Edge-based ischemic-heart-disease detection	Sensor, mobile, contextual, and medical-record data	Edge architecture and classification	Prototype / partial DT	Demonstrates feasibility of an edge-oriented Cardio Twin architecture	Limited physiological fidelity, personalization, and clinical validation

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Table 6.2: Comparative state of the art of cardiovascular Digital Twin and patient-specific virtual-heart studies (continued).

Study	Clinical objective / entity	Patient data	Modeling approach	Maturity	Evidence / main contribution	Main limitations
Gerach et al. [24]	Whole-heart electrical, mechanical, and circulatory behavior	Designed for adaptation to ECG, pressure, volume, and imaging data	Fully coupled multiscale multiphysics model	Mechanistic DT platform	Integrates electrophysiology, mechanics, contraction, and closed-loop circulation	High computational and calibration burden; limited direct patient cohort validation
Viola et al. [25]	Whole-heart multiphysics simulation and CRT scenario	Model parameters and clinical-like cardiac behavior	GPU-accelerated cardiovascular simulation	High-fidelity simulation platform	Demonstrates LBBB and resynchronization proof of concept with accelerated computation	Hours per heartbeat are not real time; patient-specific updating and prospective validation remain absent
Salvador et al. [26]	Fast whole-heart electromechanical analysis for heart failure	Simulation-derived pressure–volume traces across 43 electromechanical and hemodynamic parameters	Latent Neural ODE surrogate trained on 400 high-fidelity simulations	AI-accelerated mechanistic DT method	Enables sensitivity analysis, parameter estimation, and uncertainty quantification on a single processor	Demonstrated on simulation-derived data for one disease setting; external patient validation and domain-of-validity controls needed
Zhang et al. [21]	VT circuit prediction in arrhythmogenic right ventricular cardiomyopathy	Contrast-enhanced MRI, genotype, and clinical EP study data	Genotype-specific structural and cellular EP heart twins	Retrospective patient-specific DT	In 16 patients, reported high agreement with clinically identified VT circuit locations	Small single-center retrospective cohort; requires prospective multi-center confirmation
Hwang et al. [22]	Prediction of amiodarone efficacy after AF ablation	CT, electroanatomical maps, fibrosis, and one-year follow-up	Patient-specific left-atrial electrophysiology and virtual drug concentrations	Treatment-simulation DT	Virtual drug response stratified patients with significantly different AF recurrence outcomes	Retrospective, disease- and drug-specific; external and prospective validation needed
O’Hara et al. [23]	Preprocedural identification of post-infarction VT circuits and ablation targets	Clinical and oversampled late-gadolinium-enhanced CMR plus intraprocedural electroanatomical mapping	Patient-specific infarct geometry and rapid-pacing electrophysiological simulation	Validated offline heart twin	In six patients, higher-resolution twins better identified unique VT circuits and emergent VT than clinical-resolution twins	Very small cohort; offline workflow; imaging-resolution dependence and uncertain generalizability
Geddes et al. [27]	Noninvasive prediction of pulmonary artery pressure in heart failure	CT pulmonary angiography, catheterization, and implantable monitor data	Patient-specific geometry and 3D CFD	Calibrated proof-of-concept DT	Evaluates geometry complexity and validates pressure prediction against measured data	Small cohort; boundary conditions, imaging, and computation limit scalability
Gu et al. [29]	Heart-failure phenotyping, diagnosis, and prognosis	EHR, TTE, CMR, RHC, pressure, flow, and volume data	Mechanistic cardiovascular models plus unsupervised and supervised AI	Retrospective patient-specific model / intelligent-DT direction	343 patient twins; physiologically interpretable features improved phenotyping and prognostic modeling	Retrospective data and measurement-time mismatch; not continuously synchronized; prospective validation required

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Table 6.2: Comparative state of the art of cardiovascular Digital Twin and patient-specific virtual-heart studies (continued).

Study	Clinical objective / entity	Patient data	Modeling approach	Maturity	Evidence / main contribution	Main limitations
Qian et al. [30]	Population-scale electro-physiological mechanism discovery	Cardiac MRI, ECG, lifestyle, functional, and outcome variables	Physics-constrained cardiac twins with machine-learning-enabled personalization	Large-scale research cohort twins	3,461 UK Biobank and 359 IHD twins reveal mechanistic associations and validate selected findings	Episodic research data rather than live clinical twins; inferred parameters depend on model assumptions

6.3.6 Synthesis of the Cardiovascular State of the Art

The cardiovascular literature is more technically mature than many other eHealth domains in terms of mechanistic modeling, patient-specific geometry, and simulation of clinically meaningful interventions. It includes prospective feasibility work, treatment-response studies, large-scale cohorts, and validation against invasive measurements. Nevertheless, most systems remain calibrated snapshots rather than persistent twins. They are commonly reconstructed from imaging and episodic clinical measurements, analyzed offline, and not continuously updated as disease, treatment, and behavior change.

Three modeling directions can be distinguished. Electrophysiological twins simulate activation, repolarization, arrhythmia induction, drug effects, and ablation. Electromechanical and hemodynamic twins estimate pressures, flows, volumes, tissue mechanics, and device response. Hybrid and AI-enhanced twins use mechanistic parameters as interpretable features or train surrogates that accelerate expensive simulations. The strongest emerging systems combine these directions, but each additional modality increases calibration, identifiability, uncertainty, and validation challenges.

A recurring technical problem is non-uniqueness: different parameter combinations can reproduce similar clinical observations. A twin can therefore fit an ECG or pressure curve while representing the wrong underlying physiology. Credible cardiovascular twins need uncertainty distributions, sensitivity analysis, independent measurements, and validation against outcomes not used for calibration. This is especially important when the twin is used to recommend an ablation target, predict treatment response, or estimate an invasive marker.

6.4 Cross-Domain Comparison and Research Gaps

The two states of the art reveal a gap between architectural completeness and physiological fidelity. General eHealth research provides layered architectures, interoperability principles, governance requirements, and broad service taxonomies. Cardiovascular research provides detailed mechanistic and patient-specific models with stronger physiological interpretation. Few studies combine both strengths in a single system that is interoperable, longitudinal, dynamically updated, clinically validated, and usable in routine care.

The principal research gaps are:

- **Definition and maturity:** systems continue to use the Digital Twin label for static models, offline simulations, and one-way data pipelines.
- **Longitudinal synchronization:** most patient models are calibrated from episodic data and do not evolve continuously with the patient.
- **Multimodal integration:** imaging, biosignals, EHR data, laboratory values, wearables, and patient-reported data remain difficult to align semantically and temporally.
- **Identifiability and uncertainty:** multiple parameter sets may explain the same measurements, while uncertainty is often incompletely reported.

- **Hybrid-model validation:** AI can accelerate or personalize mechanistic models, but errors from data-driven and physics-based components must be evaluated jointly.
- **Clinical evidence:** retrospective and proof-of-concept studies dominate; external, prospective, multi-center, and workflow-level validation remain limited.
- **Adaptive-model governance:** continuous learning introduces drift, revalidation, auditability, and regulatory problems.
- **Interoperability and provenance:** many prototypes do not demonstrate robust integration with clinical information systems or traceable data lifecycles.
- **Human factors and responsibility:** the effect of twin outputs on clinician behavior, patient understanding, alert burden, accountability, and trust is underexplored.
- **Equity and access:** high-fidelity models may depend on specialized imaging, invasive tests, and high-performance computing that are not equally available.

A future clinically credible eHealth Digital Twin should therefore be defined not only by model sophistication but by a complete evidence chain: verified software, traceable and interoperable data, patient-specific calibration, uncertainty-aware prediction, external and prospective validation, human-supervised services, and controlled feedback. Cardiovascular studies offer strong examples of physiological modeling and treatment simulation, while general eHealth studies provide the architectural and governance foundation required to translate these capabilities into a broader healthcare system.

6.5 Conclusion

This chapter developed two distinct but connected states of the art. The first showed how healthcare Digital Twins extend from patient-specific medicine to hospital processes, public health, and AI-enabled services. The second examined cardiovascular Digital Twins in depth, including arrhythmia risk, ablation planning, virtual drug testing, whole-heart modeling, heart-failure hemodynamics, interpretable AI, and population-scale twinning. The comparative analyses demonstrate that the field has progressed beyond purely conceptual discussion, but fully connected and clinically validated patient twins remain uncommon. These findings provide the evidence base for the architecture and practical specialization developed in the following parts of the thesis.

Part IV

Proposed Method

CHAPTER 7

PROPOSED ARCHITECTURE FOR A DIGITAL TWIN-BASED EHEALTH SYSTEM

7.1 Architecture Rationale

Based on the literature review, a Digital Twin-based eHealth system should not be limited to a data dashboard or a predictive algorithm. It should be organized as a layered system that connects the physical healthcare environment to a virtual representation, processes heterogeneous data, provides simulation and analysis services, and ensures security, privacy, and interoperability.

The proposed architecture is general and can be adapted to different eHealth scenarios such as remote monitoring, chronic disease management, hospital workflow optimization, rehabilitation support, wellness tracking, or clinical decision support. It is not designed for one specific disease or one specific project.

7.2 Overview of the Proposed Architecture

The proposed architecture is composed of seven layers: physical layer, data acquisition layer, interoperability and preprocessing layer, data management layer, virtual twin layer, intelligence and simulation layer, and service layer. Security, privacy, and governance are considered as cross-cutting concerns across all layers.

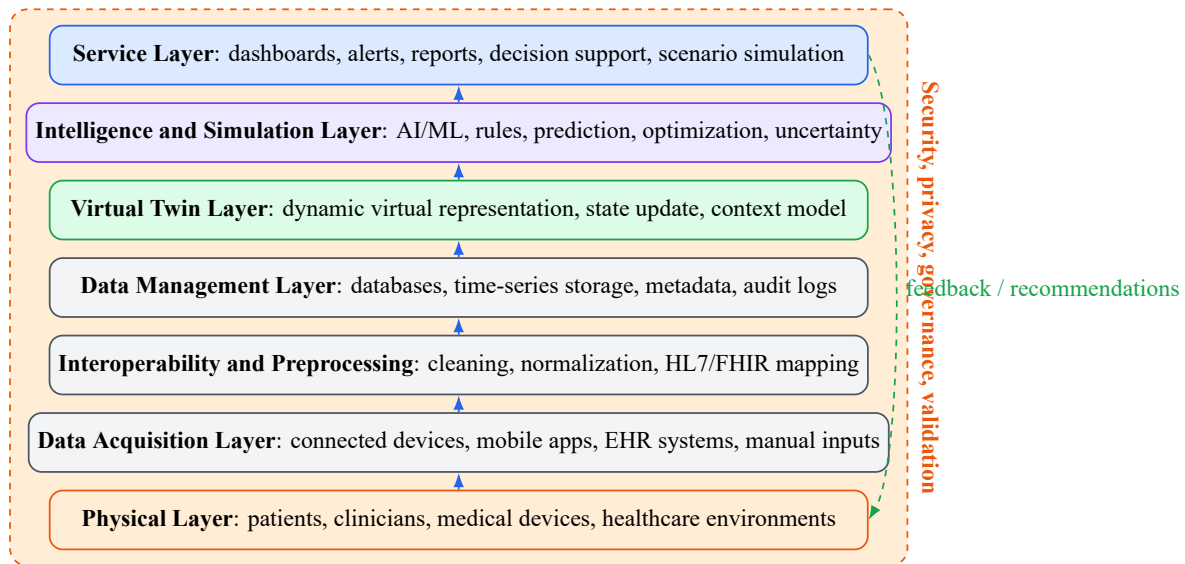


Figure 7.1: Proposed multi-layer architecture for a Digital Twin-based eHealth system.

7.3 Physical Layer

The physical layer includes real-world entities involved in the eHealth ecosystem. These may include patients, healthcare professionals, medical devices, wearable sensors, mobile phones, hospital equipment, and healthcare environments. The role of this layer is to generate real-world data and receive possible feedback or recommendations from the digital services.

7.4 Data Acquisition Layer

The data acquisition layer collects information from connected devices, mobile applications, clinical systems, and manual inputs. Data may be continuous, periodic, or event-based. Examples include physiological measurements, activity data, medication information, symptoms, clinical notes, laboratory results, and operational data.

7.5 Interoperability and Preprocessing Layer

This layer prepares data for use by the Digital Twin. It handles data cleaning, normalization, timestamp alignment, missing value management, anomaly detection, and format conversion. It also supports interoperability standards such as HL7 and FHIR to connect the Digital Twin with healthcare information systems.

7.6 Data Management Layer

The data management layer stores historical, real-time, and contextual information. It may include relational databases, time-series databases, clinical repositories, data warehouses, and metadata catalogs. Data governance rules must define who can access data, how long data are stored, and how data are anonymized or pseudonymized when needed.

7.7 Virtual Twin Layer

The virtual twin layer represents the current state of the physical entity or healthcare process. Depending on the application, the virtual twin may represent a patient profile, a disease trajectory, a medical device, a care pathway, a hospital service, or a public health process. The virtual twin should be continuously updated when new valid data become available.

7.8 Intelligence and Simulation Layer

This layer provides analytical and predictive capabilities. It may include rule-based reasoning, machine learning models, deep learning models, statistical models, mechanistic simulations, optimization algorithms, and uncertainty estimation. The purpose is to transform the virtual representation into useful knowledge for monitoring, prediction, scenario analysis, and decision support.

7.9 Service Layer

The service layer exposes the functions of the Digital Twin to users and systems. Possible services include dashboards, alerts, reports, risk assessment, scenario simulation, remote monitoring, workflow optimization, and decision support. These services must be designed according to user needs, clinical safety, explainability, and usability requirements.

7.10 Security, Privacy, and Governance Layer

Security and privacy are cross-cutting concerns. The system should include authentication, authorization, encryption, audit logs, secure APIs, role-based access control, data minimization, consent management, and governance policies. Ethical oversight is also required to ensure fairness, transparency, and responsible use of the Digital Twin.

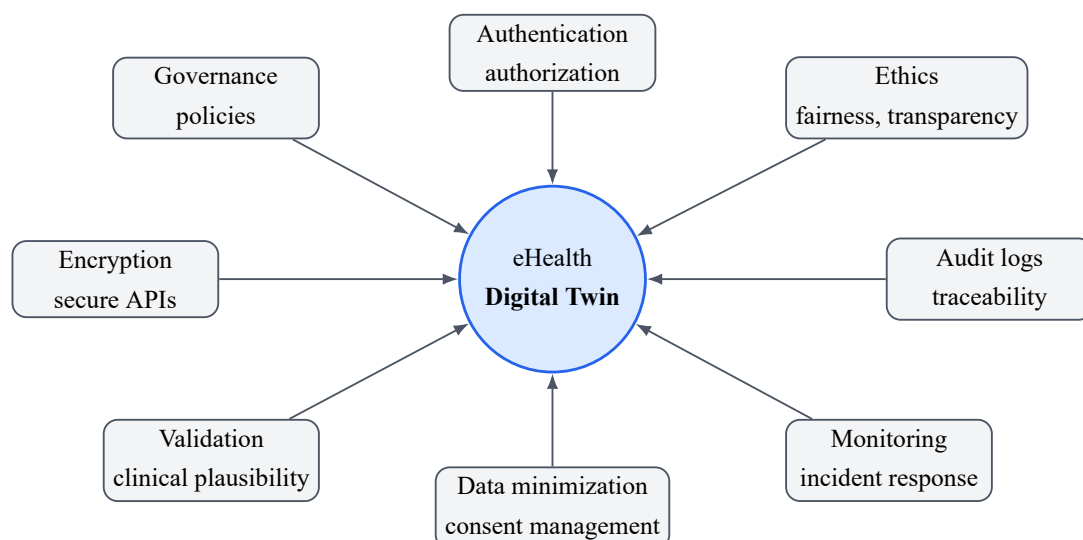


Figure 7.2: Security, privacy, and governance concerns around the eHealth Digital Twin.

7.11 Data Flow

Figure 7.3 illustrates the general data flow. Real-world data are collected from the physical layer, transmitted through acquisition modules, preprocessed and standardized, stored in the data layer, and used to update the virtual twin. The intelligence and simulation layer generates insights that are exposed through services. Feedback may then be provided to patients, clinicians, or healthcare systems depending on the use case.

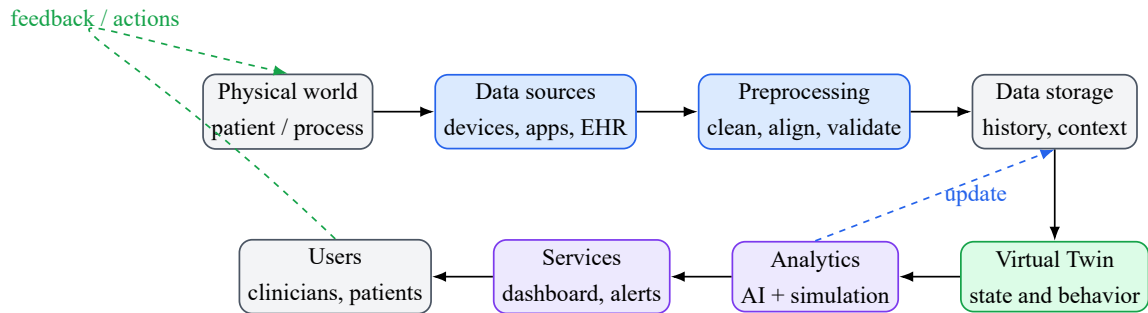


Figure 7.3: General data flow in a Digital Twin-based eHealth system.

7.12 General Use Case Scenario

A general eHealth use case can be described as follows. A patient uses connected devices and a mobile application to collect health-related data at home. These data are transmitted to the eHealth platform, cleaned, standardized, and stored. The virtual twin is updated with the new measurements and historical context. The intelligence layer analyzes trends, detects anomalies, estimates risk, and simulates possible scenarios. The service layer displays a dashboard, generates alerts when necessary, and produces a report for healthcare professionals.

This scenario remains general and can be adapted to different domains such as chronic disease follow-up, elderly care, rehabilitation, mental health monitoring, wellness tracking, hospital-at-home programs, and clinical workflow optimization.

7.13 Evaluation Plan

The proposed architecture can be evaluated using several criteria:

- **Functional evaluation:** ability to collect, store, update, analyze, and display data.
- **Interoperability evaluation:** compatibility with standards and existing healthcare systems.
- **Performance evaluation:** latency, scalability, and reliability.
- **Security evaluation:** authentication, authorization, encryption, and auditability.
- **Usability evaluation:** clarity of dashboards and usefulness for users.
- **Clinical plausibility:** relevance and safety of generated outputs.
- **Ethical evaluation:** fairness, transparency, consent, and data governance.

7.14 Conclusion

This chapter proposed a general conceptual architecture for Digital Twin-based eHealth systems. The architecture is designed to be reusable, modular, secure, interoperable, and adaptable to multiple healthcare scenarios.

Part V

Conclusion

CHAPTER 8

GENERAL CONCLUSION AND FUTURE WORK

8.1 General Conclusion

This thesis studied the Digital Twin paradigm in the context of eHealth. It showed that Digital Twin technology can support the transition from fragmented digital health applications toward dynamic, personalized, and service-oriented healthcare systems. A Digital Twin is not only a digital copy or a visualization tool; it is a connected virtual representation that uses data, models, and services to support monitoring, simulation, prediction, and decision-making.

The literature review showed that Digital Twins for Health are receiving growing attention, but the field remains emerging. Existing works provide valuable definitions, classifications, architectures, and use cases. However, many systems are still conceptual, prototype-based, or limited to specific scenarios. The implementation of reliable eHealth Digital Twins requires interoperability, security, privacy, clinical validation, explainability, and ethical governance.

To address these needs, this thesis proposed a general multi-layer architecture for Digital Twin-based eHealth systems. The architecture includes physical, acquisition, interoperability, data management, virtual twin, intelligence, simulation, and service layers, with security and privacy as cross-cutting requirements.

8.2 Limitations

This thesis is primarily an academic state-of-the-art and conceptual architecture study. It does not provide a full clinical deployment or a validated medical product. The proposed architecture must be adapted and evaluated according to a specific healthcare use case, available data sources, regulatory constraints, and user requirements.

8.3 Future Work

Future work may focus on implementing and evaluating the proposed architecture in real or semi-real eHealth scenarios. Important directions include:

- integration with standardized EHR systems using HL7/FHIR;
- development of secure data acquisition pipelines from connected medical devices;
- design of explainable AI and simulation modules;
- validation using realistic healthcare datasets;
- evaluation of usability with healthcare professionals and patients;
- strengthening privacy, consent management, and ethical governance;
- extension toward multi-twin ecosystems connecting patients, devices, workflows, and healthcare services.

8.4 Final Remarks

Digital Twins for eHealth represent a promising research direction at the intersection of healthcare, computer science, data engineering, artificial intelligence, and systems modeling. Their success will depend not only on technical performance, but also on trust, interoperability, clinical relevance, and responsible design.

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APPENDIX A

APPENDIX: FIGURE AND TABLE PLACEHOLDERS

This appendix lists figure ideas that can be added later if the thesis is expanded.

- Detailed PRISMA diagram with final numbers after paper selection.
- Timeline of Digital Twin evolution from industrial systems to eHealth.
- Detailed comparison of Digital Twin architectures in healthcare.
- Data model showing Patient, Observation, Device, Encounter, and Service entities.
- Security threat model for Digital Twin-based eHealth systems.
- Evaluation framework diagram linking technical, clinical, ethical, and usability criteria.

Figure placeholder

Add the final PRISMA diagram here after selecting the exact number of papers.

Figure placeholder

Add a refined architecture diagram if the supervisor asks for more technical detail.